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EXPERIMENT 5

Exp No:5 Data Collection and Initial Exploration

AIM:
To collect, load, and perform initial exploration of the diabetes dataset.

Materials Needed:

 Diabetes dataset (available from sources such as UCI Machine Learning Repository or Kaggle)
 Python with libraries: pandas, numpy, matplotlib, seaborn

Procedure and Code:

```
import pandas as pd

db= pd.read_csv("/content/Diabetes_prediction.csv")

# Display the first few rows

print(db.head())

print(db.info())

print(db.describe())

import matplotlib.pyplot as plt

import seaborn as sns

db.hist(bins=50,figsize=(20,15))

plt.show()

sns.pairplot(db)

plt.show()
```

OUTPUT:

```
Pregnancies     Glucose  BloodPressure  SkinThickness     Insulin  \
0           2  115.863387      56.410731      24.336736   94.385783
1           2   92.490122      70.615520      23.443591  138.652426
2           1   88.141469      63.262618      23.404364  149.358082
3           2  108.453101      67.793632      20.751580  108.751638
4           1  127.849443      94.725685      22.603078   25.269987

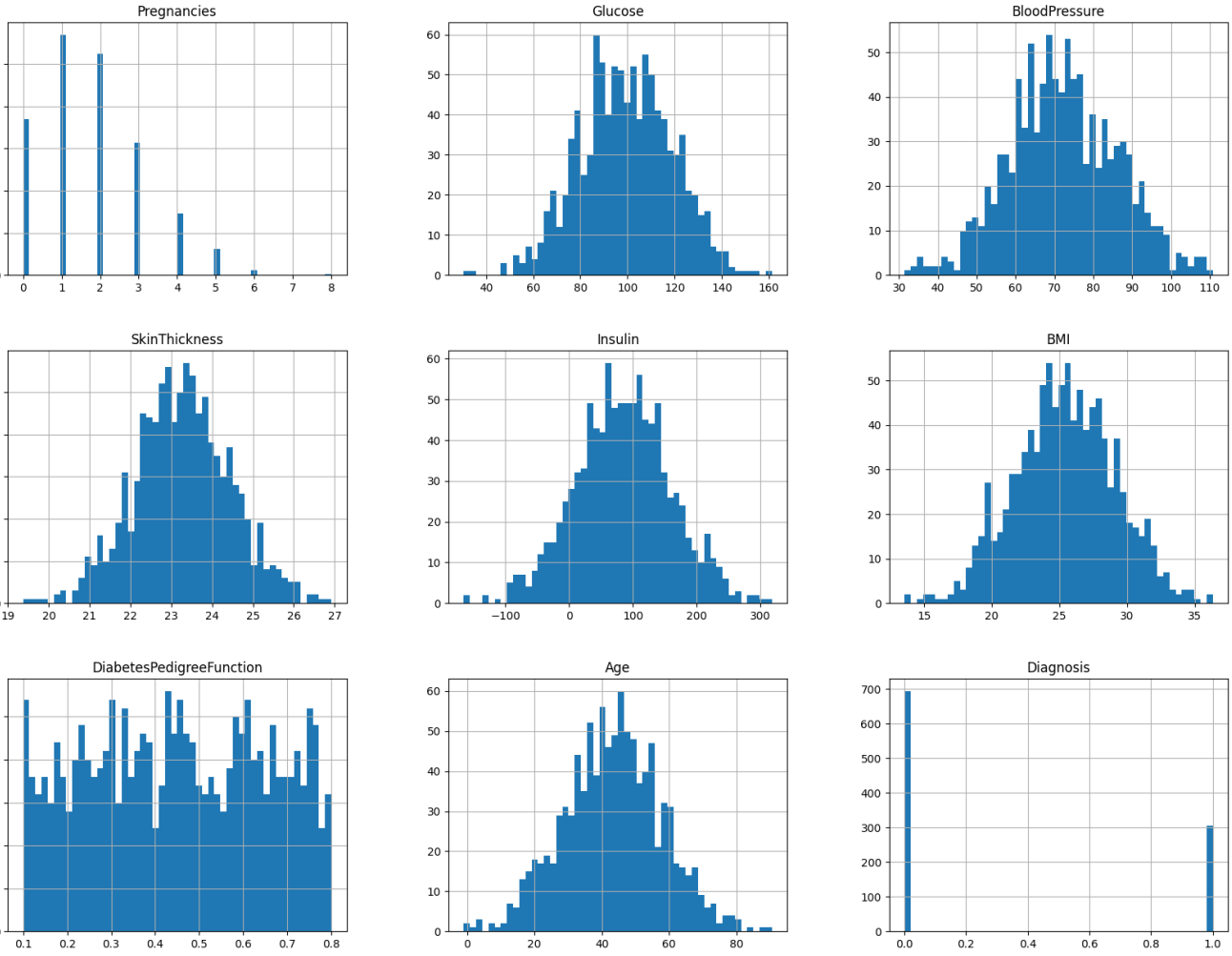
        BMI  DiabetesPedigreeFunction        Age  Diagnosis
0  26.455940                 0.272682  20.100494          0
1  23.910167                 0.665160  44.912281          0
2  21.948250                 0.676022  48.247873          1
3  24.209304                 0.289636  42.749868          0
4  32.997477                 0.601315  32.797789          0
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1000 entries, 0 to 999
Data columns (total 9 columns):
 #   Column                    Non-Null Count  Dtype
---  ------                    --------------  -----
 0   Pregnancies               1000 non-null   int64
 1   Glucose                   1000 non-null   float64
 2   BloodPressure             1000 non-null   float64
 3   SkinThickness             1000 non-null   float64
 4   Insulin                   1000 non-null   float64
 5   BMI                       1000 non-null   float64
 6   DiabetesPedigreeFunction  1000 non-null   float64
 7   Age                       1000 non-null   float64
 8   Diagnosis                 1000 non-null   int64
dtypes: float64(7), int64(2)
```

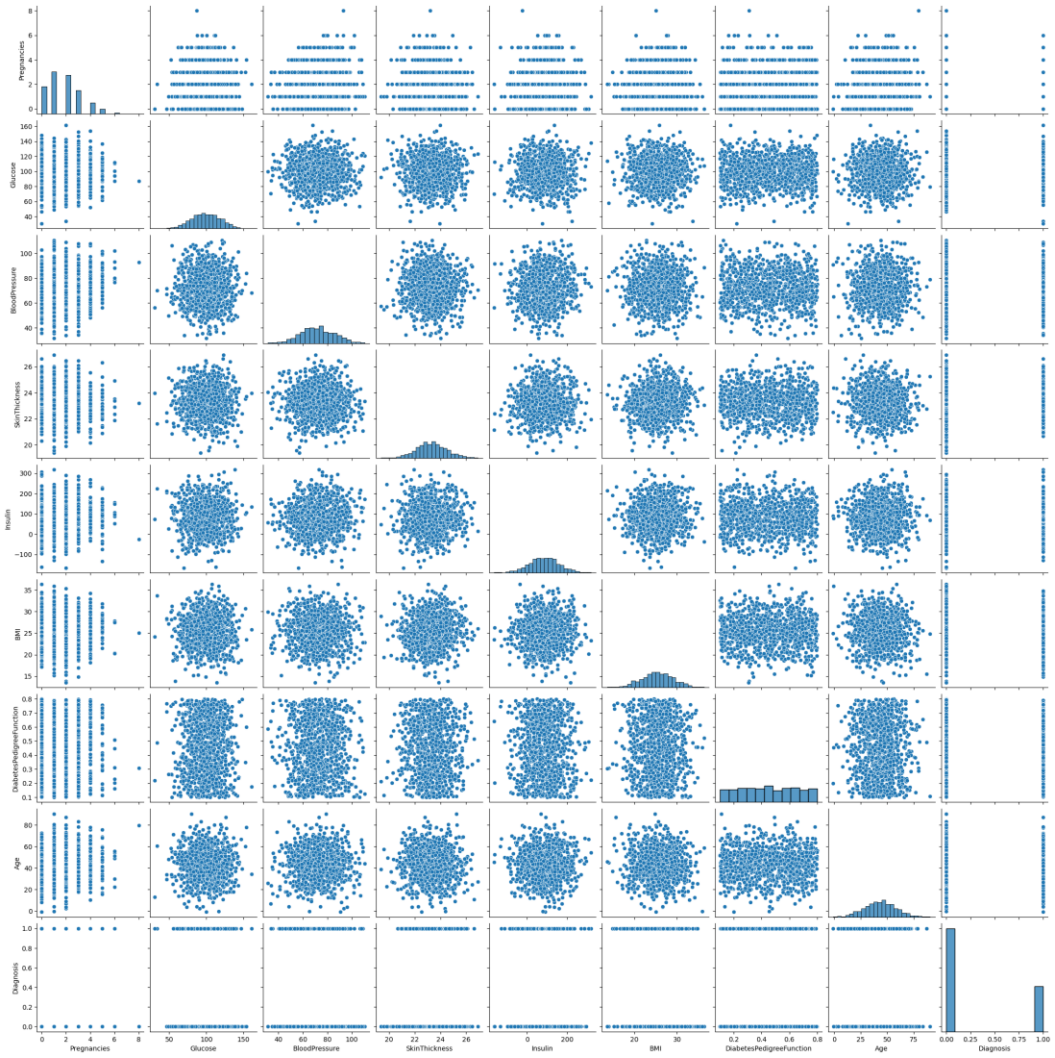
memory usage: 70.4 KB

None

	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin \
count	1000.000000	1000.000000	1000.000000	1000.000000	1000.000000
mean	1.771000	99.440607	72.179837	23.278316	84.582679
std	1.354398	19.470730	13.882017	1.173807	74.872733
min	0.000000	30.571402	31.401487	19.369987	-165.310033
25%	1.000000	86.145927	62.795447	22.501591	35.076535
50%	2.000000	99.458362	71.909588	23.275225	84.442232
75%	3.000000	113.264556	82.082660	24.052022	134.267842
max	8.000000	161.238939	110.723715	26.917654	317.701852

	BMI	DiabetesPedigreeFunction	Age	Diagnosis
count	1000.000000	1000.000000	1000.000000	1000.00000
mean	25.433600	0.449383	43.281798	0.30600
std	3.690223	0.199334	14.465398	0.46106
min	13.548818	0.100037	-0.979804	0.00000
25%	23.022715	0.283376	33.518451	0.00000
50%	25.455649	0.448219	43.634273	0.00000
75%	27.972184	0.619158	53.098446	1.00000
max	36.324598	0.799654	90.573782	1.00000


Pregnancies
Glucose
BloodPressure
SkinThickness
Insulin
BMI
DiabetesPedigreeFunction
Age
Diagnosis



RESULT:

The diabetes dataset was successfully loaded and initially explored. The first few records, dataset structure, and summary statistics were displayed. Histogram showed the distribution of each feature, and the pair plot showed the relationship between the variables**.**